STUDENT RECORD SYSTEM

BRANCH COVERAGE REPORT

Statement and interpretation of conditional decision-path coverage

Document	TEST-BR-04
Project	Student Record System
Course	Software Engineering Mini Project
Team	Team 11
Implementation	C++17 application with SQLite persistence
Report basis	Recorded project test results and coverage.xml evidence available in the project conversation
Report date	10 November 2026

Prepared as a final testing report for the project documentation package.

1. Objective

This report presents branch coverage from the supplied GCOVR 7.2 coverage.xml artifact. Branch coverage measures how many instrumented decision outcomes were exercised by the test workload.

2. Measurement Tools and Evidence

Item	Recorded value
Instrumentation	GCC coverage instrumentation (--coverage)
Report generator	GCOVR 7.2
Evidence file	coverage.xml supplied from the GitHub Actions coverage artifact
Coverage package	src
Coverage timestamp in XML	1794292553 (Unix timestamp recorded in the XML)

3. Overall Branch Coverage

36.03%	838 / 2,326	1,488
Branch coverage	Branches executed / valid	Branches not executed

Calculation: 838 covered branches ÷ 2,326 valid branches × 100 = 36.0275%, reported as 36.03%.

4. Interpretation

• 838 branch outcomes were executed by the measured test workload.

• 1,488 branch outcomes were not executed by that workload.

• Branch coverage is lower than line coverage because executing a line containing a condition does not necessarily exercise every possible outcome of that condition.

• The uncovered outcomes indicate decision paths that the current workload did not reach; they do not independently prove a functional defect.

5. Coverage Scope and Limitations

• The supplied XML identifies the coverage package as src.

• The result is based on the supplied coverage.xml artifact, not a fresh local rerun.

• Branch coverage depends on the source files included by the configured GCOVR filters and the paths exercised in the coverage run.

• This metric does not measure the quality of assertions or prove that all SRS requirements have been validated.

• The XML is the source of truth for the aggregate branch counts in this report.

6. Result

The supplied coverage artifact reports 36.03% branch coverage (838 of 2,326 valid branches).

7. Conclusion and Improvement Opportunity

The recorded workload exercises 36.03% of the valid branch outcomes in the measured source scope. To increase this figure, additional tests should target the unexecuted true/false outcomes of conditionals, validation failures, permission-denial paths, empty-result paths, and error-handling branches. Such tests should be added only where the corresponding source behavior exists and can be verified.